10/22/2026

LAB NO 07

Operating System Lab

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Class: BSCS F'24 B

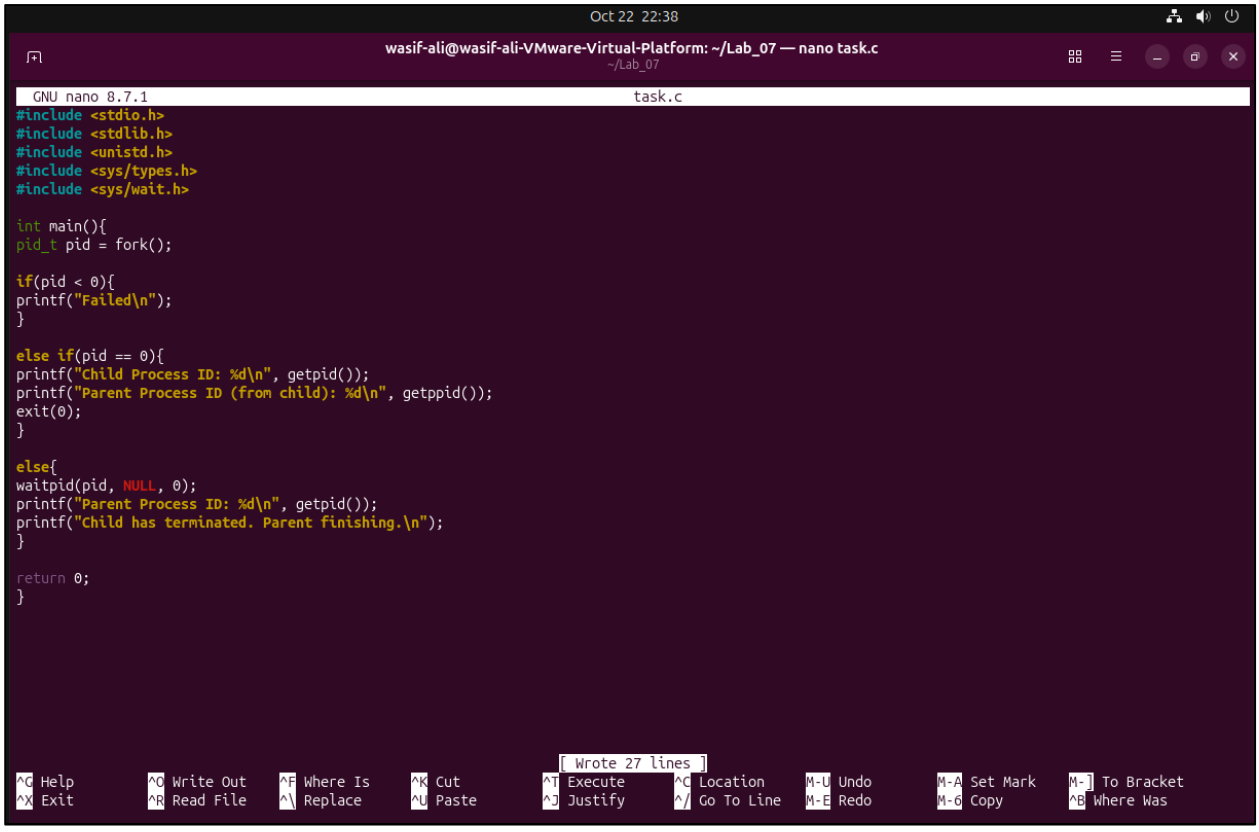
LAB TASK

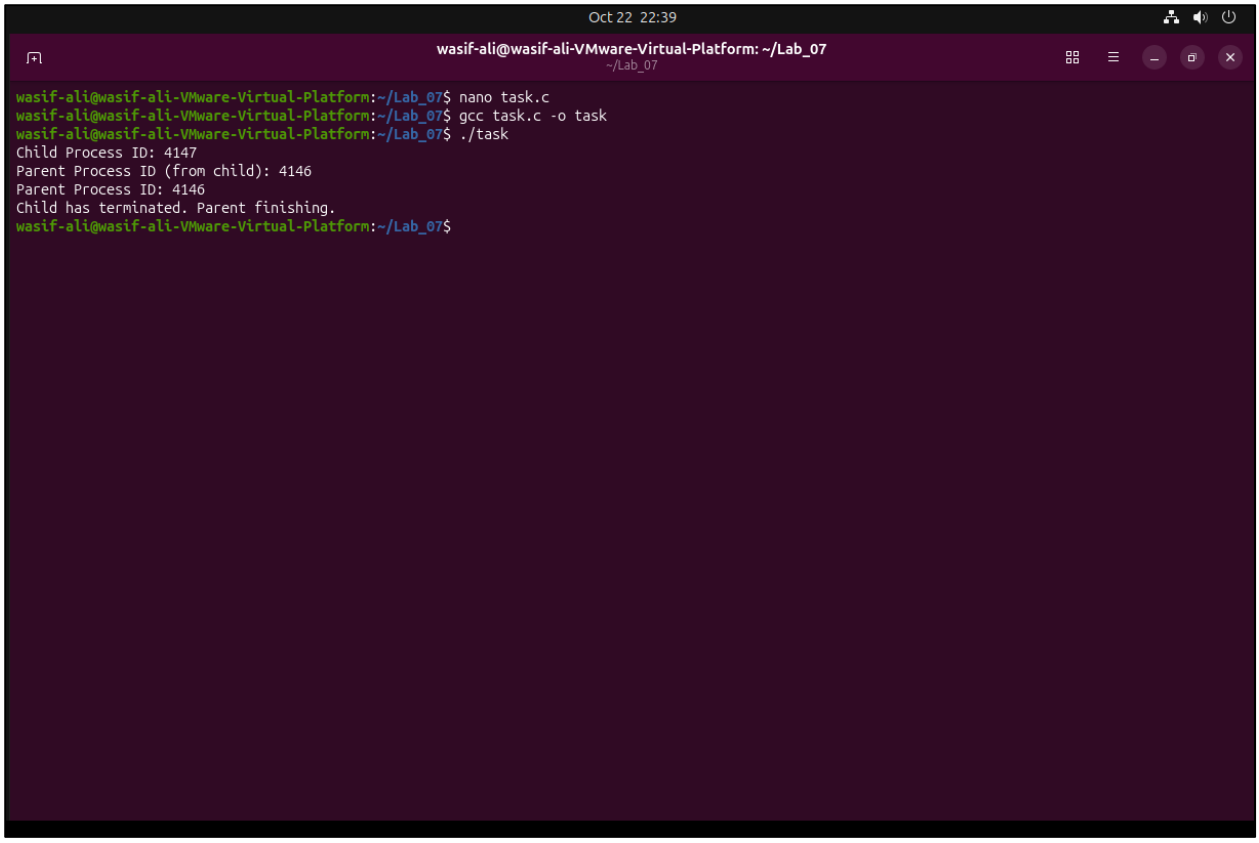
Lab Task: Process Creation and Management

Problem Statement:

Solve the process management problem by developing a C program that uses fork() to create a child process. Use getpid() and getppid() to display the process IDs of the parent and child. Demonstrate the use of waitpid() to synchronize process execution and exit() to terminate the child process. Show the output of the program and explain how the parent and child processes execute independently.

Code/Commands:





In this final task, we combined everything we learned about process management. We created a C program that uses fork() to spawn a child process. Inside the child process, we displayed its own PID and its parent's PID using getpid() and getppid(), and then safely terminated it using exit(0). In the parent process, we used waitpid() to ensure the parent pauses and waits specifically for this child to finish before printing its own PID. Parent and child processes execute independently because they run in separate memory spaces; the operating system treats them as distinct tasks, but by using waitpid(), we successfully forced them to synchronize their execution.